NOVACART DATA CLEANING

1. NOVACART INFO – Raw Data

I checked the original dataset to identify missing values, blank cells, duplicates, and formatting issues.

I used filters to find missing values in different columns and created a separate CALC sheet to analyse the data.

I organized the cleaned data into three separate sheets: Customer Data, Product Data, and Order Data.

2. CALC Sheet

I created a CALC sheet to analyse the dataset and find suitable values to fill the missing data.

I used MODE and AVERAGE to calculate the following:

- Mode of Order ID – 1044
- Mode of Quantity – 1
- Average Unit Price – 38.84
- Average Discount – 5.5
- Mode of Returned – 0

I also counted the products and created a PivotTable to find the most repeated product in each category.

The most repeated products were:

- Apparel – Joggers
- Beauty – Vitamin C Serum
- Electronics – Wireless Earbuds
- Home – LED Lamp
- Sports – Water Bottle
- Toys – RC Car

I used these results to fill missing product names based on their categories.

I also checked the sales channels and found that Organic had the highest count of 272.

After filling the missing values, I filtered the dataset again to check for remaining blanks.

3. Customer Data Sheet

I created a separate Customer Data sheet with Customer ID, Customer Full Name, First Name, Last Name, Email Address, and Segment.

- I used UNIQUE to get customer names without duplicates.
- I used XLOOKUP to get Customer IDs from the original dataset.
- I used TEXTBEFORE to separate the first names.
- I used TEXTAFTER to separate the last names.
- I used LOWER to convert the first and last names into lowercase.

- I used CONCATENATE to create email addresses using the first and last names.
- I used XLOOKUP to get the customer segments based on Customer IDs.
- I arranged and formatted the customer details.

4. Product Data Sheet

I created a separate Product Data sheet with Product ID, Product Name, and Category.

- I used UNIQUE to get product names without duplicates.
- I used SORT to arrange the product names alphabetically.
- I used XLOOKUP to get the Product IDs based on Product Names.
- I used VLOOKUP to get the Categories based on Product Names.
- I checked the Product IDs and Categories with the original dataset.
- I arranged the product information in a separate table.

5. Order Data Sheet

I created a separate Order Data sheet with Order Date, Order ID, Customer ID, Product ID, Channel, Quantity, Unit Price, Discount %, Line Total, and Returned.

- I used VLOOKUP to get Order Dates based on Order IDs.
- I changed the Order Date format to DD-MM-YYYY.
- I used VLOOKUP to get Customer IDs from the original dataset.
- I filled the Product IDs using the product information.
- I used VLOOKUP to get the sales channels.
- I filled the Quantity, Unit Price, and Discount % columns.
- I calculated Line Total using Quantity, Unit Price, and Discount %.
- I used VLOOKUP to get the Returned values.
- I formatted Unit Price and Line Total as currency.
- I arranged the columns and checked the data.

6. Data Cleaning and Formatting

I used filters to identify missing values and blank cells.

I used the CALC sheet results to fill missing values where needed.

I used the IF function to fill missing Product Names based on their Categories.

I used lookup functions to fill missing Product IDs and Categories based on the available product information.

I also checked the customer and product details, formatted the dates, and adjusted the number and currency formats.

After filling the missing values, I filtered the columns again to check for remaining blanks.

7. Excel Functions and Features Used

- UNIQUE
- SORT
- FILTER

- XLOOKUP
- VLOOKUP
- IF
- MODE
- AVERAGE
- TEXTBEFORE
- TEXTAFTER
- LOWER
- CONCATENATE
- PivotTable
- Excel Filters
- Date and Currency Formatting